ON LOCALIZATION PHENOMENA IN PLASTICITY MODELS

BERNARDO ARDINI

Abstract.

Contents

1. Introduction and notation

2. The tensile test at small deformations

In this section we develop a one dimensional model for an elastoplastic bar at small deformations. We prove the existence of solutions and we study the boundary value problem that describe tensile test. We show that the model is not able to predict the onset of necking.

Date: 30 March 2026.

2.1. **Multidimensional kinematics.** Let $\Omega = (0, L) \times \Sigma$ with $L > 0$ and $\Sigma \subset \mathbb{R}^{d-1}$ be the reference configuration of the cylindrical bar. We denote the generic point in Ω as (x, y) with $x \in (0, L)$ and $y \in \Sigma$. We assume a macroscopic displacement of the form

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} u_1(x) \\ u_2(x)y \end{pmatrix}$$

where u_1 and u_2 are functions from $(0, L)$ to $\mathbb{R}$. This assumption means that the sections of the cylindrical bar remain flat and can translate and expand uniformly. The corresponding displacement gradient is

$$H = \begin{pmatrix} u_1' & 0 \\ u_2'y & u_2 I \end{pmatrix}.$$

The plastic distortion is assumed to be of the form

$$H_p = \begin{pmatrix} p & 0 \\ 0 & -\frac{p}{d-1} I \end{pmatrix}$$

where $p\colon (0, L) \to \mathbb{R}$. This choice implies plastic incompressibility. The elastic distortion is then given by

$$H_e = H - H_p = \begin{pmatrix} u_1' - p & 0 \\ u_2'y & \left(u_2 + \frac{p}{d-1}\right) I \end{pmatrix}.$$

Then the elastic strain is

$$E_e = \begin{pmatrix} u_1' - p & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & \left(u_2 + \frac{p}{d-1}\right) I \end{pmatrix} = \begin{pmatrix} e_1 & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & e_2 I \end{pmatrix}$$

where we have introduced $e_1 = u_1' - p$ and $e_2 = u_2 + (d-1)^{-1}p$. Assuming an isotropic elastic response the elastic free energy density is given by

$$\psi_e = \frac{1}{2}\kappa(\operatorname{tr} E_e)^2 + \mu|\operatorname{dev} E_e|^2$$

where $\kappa \geq 0$ and $\mu \geq 0$ are the bulk and shear modulus respectively. We notice that

$$\operatorname{tr} E_e = e_1 + (d-1)e_2 \quad \text{and} \quad \operatorname{dev} E_e = \begin{pmatrix} \frac{d-1}{d}(e_1 - e_2) & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & -\frac{1}{d}(e_1 - e_2) I \end{pmatrix}.$$

Then the elastic free energy density can be written as

$$\psi_e = \frac{1}{2}\kappa \left(e_1 + (d-1)e_2\right)^2 + \frac{d-1}{d}\mu(e_1 - e_2)^2 + \frac{1}{2}|y|^2\mu|u_2'|^2.$$

We conclude that the elastic free energy per unit axial length is

$$\int_\Sigma \psi_e dy = \frac{1}{2}A\kappa \left(e_1 + (d-1)e_2\right)^2 + \frac{d-1}{d}A\mu(e_1 - e_2)^2 + \frac{1}{2}J\mu|u_2'|^2$$

where

$$A = \int_\Sigma dy \quad \text{and} \quad J = \int_\Sigma |y|^2 dy$$

are the polar and the polar second order moment of area.

2.2. **One dimensional kinematics.**

3. The tensile test at large deformations

4. Shear band localization in cohesive materials

Appendix A. Modeling in continuum mechanics

Appendix B. Tools from functional analysis

B.1. **Other results.**

Proposition B.1. (Gronwall inequality) Let $z \in L^\infty((0,T))$. Let $C \in \mathbb{R}$ and α, β in $L^1((0,T),[0,\infty))$ such that

$$y(t) \leq C + \int_0^t (\alpha(s)y(s) + \beta(s))\, ds \quad \text{for a.e. } t \in (0,T).$$

Then

$$z(t) \leq e^{\int_0^t \alpha(s)ds} \left(C + \int_0^t \beta(s) e^{-\int_0^s \alpha(r)dr} \right) \quad \text{for a.e. } t \in (0,T).$$

Proof. Let

$$Z(t) = C + \int_0^t (\alpha(s)z(s) + \beta(s))\, ds$$

so that $z(t) \leq Z(t)$. Clearly $Z \in \mathrm{AC}([0,T])$. It holds that

$$\dot{Z}(t) = \alpha(t)z(t) + \beta(t) \leq \alpha Z(t) + \beta(t) \quad \text{for a.e. } t \in (0,T).$$

Multiplying by $e^{-\int_0^t \alpha(s)ds}$ one gets

$$\frac{d}{dt}\left(Z(t) e^{-\int_0^t \alpha(s)ds} \right) \leq \beta(t) e^{-\int_0^t \alpha(s)ds}$$

and integrating on $(0,t)$

$$Z(t) e^{-\int_0^t \alpha(s)ds} \leq C + \int_0^t \beta(s) e^{-\int_0^s \alpha(r)dr} ds$$

Then

$$z(t) \leq Z(t) \leq e^{\int_0^t \alpha(s)ds} \left(C + \int_0^t \beta(s) e^{-\int_0^s \alpha(r)dr} \right) \quad \text{for a.e. } t \in (0,T).$$

□

B.2. **Nonlinear analysis.**

Proposition B.2. Let V a Banach space and let $f \in C^1(V)$ and $K \subset V$ compact. Then f is Lipschitz continuous on K.

Proof. By contradiction let u_n and v_n two sequences in K such that

$$\frac{|F(u_n) - F(v_n)|}{\|u_n - v_n\|} \geq n.$$

By compactness of K up to extracting subsequences we can assume that $u_n \to u$ and $v_n \to v$. If $u \neq v$ then

$$\frac{|F(u_n) - F(v_n)|}{\|u_n - v_n\|} \to \frac{|F(u) - F(v)|}{\|u - v\|} < \infty$$

which is a contradiction. If $u = v$ then

$$\frac{|F(u_n) - F(v_n)|}{\|u_n - v_n\|} = \frac{|\langle DF(u), u_n - v_n\rangle + o(\|u_n - v_n\|)|}{\|u_n - v_n\|} \le \|DF(u)\| + o(1) \to \|DF(u)\| < \infty$$

which is a contradiction. □

B.3. **Convex analysis.**

Example B.3. Let $f(v) = \frac{1}{2}\|v\|^2$. Then $f^*(\xi) = \frac{1}{2}\|\xi\|^2$. Indeed

$$\langle \xi, v\rangle - \frac{1}{2}\|v\|^2 \le \frac{2\|\xi\|\|v\| - \|v\|^2}{2} \le \frac{\|\xi\|^2}{2}.$$

This implies $f^*(\xi) \le \frac{\|\xi\|^2}{2}$. To get the opposite inequality, fix $\epsilon > 0$ and let $v \in V$ such that $\|v\| = \|\xi\|$ and $\langle \xi, v\rangle \ge \|\xi\|^2 - \epsilon$. Then

$$f^*(\xi) \ge \|\xi\|^2 - \epsilon - \frac{1}{2}\|\xi\|^2 \ge \frac{1}{2}\|\xi\|^2 - \epsilon.$$

By arbitrariness of ϵ we conclude.

Let f and g from V to $(-\infty, \infty]$. Their inf-convolution $f \square g \colon V \to (-\infty, \infty]$ is

$$f \square g(v) = \inf_{w \in V} (f(w) + g(v - w)).$$

Proposition B.4. (inf-convolution duality formula) Let f and g from V to $(-\infty, \infty]$. Then

(i) $(f \square g)^* = f^* + g^*$
(ii) $(f + g)^* \le f^* \square g^*$
(iii) if there exists $v \in \operatorname{dom} f \cap \operatorname{dom} g$ and f is continuous at v then $(f + g)^* = f^* \square g^*$.

Proof. Step 1. It holds that

$$\begin{aligned}
(f \square g)^* &= \sup_{v \in V} \left(\langle \xi, v\rangle - \inf_{w \in V} (f(w) + g(v - w)) \right) = \\
&= \sup_{v \in V} \sup_{w \in V} (\langle \xi, w\rangle - f(w) + \langle \xi, v - w\rangle - g(v - w)) = \\
&= \sup_{w \in V} \sup_{v \in V} (\langle \xi, w\rangle - f(w) + \langle \xi, v - w\rangle - g(v - w)) = \\
&= \sup_{w \in V} (\langle \xi, w\rangle - f(w) + g^*(\xi)) = f^*(\xi) + g^*(\xi).
\end{aligned}$$

This establish (i).

Step 2. By the Young–Fenchel inequality

$$f^*(\zeta) + g^*(\xi - \zeta) \ge \langle \xi, v\rangle - f(v) - g(v).$$

Taking the supremum over $v \in V$

$$f^*(\zeta) + g^*(\xi - \zeta) \ge (f + g)^*$$

and taking the infimum over $\zeta \in V$

$$f^* \square g^* \ge (f + g)^*.$$

This proves (ii).

Step 3. Now we prove (iii). If $(f + g)^*(\xi) = \infty$ then (iii) is satisfied at ξ. Then suppose that $(f + g)^*(\xi) \in [-\infty, \infty)$. Since $\operatorname{dom} f \cap \operatorname{dom} g \ne \emptyset$ then actually $(f + g)^*(\xi) = c \in (-\infty, \infty)$. Let

$$A = \{(a, v) \in \mathbb{R} \times V \mid a \le \langle \xi, v\rangle - g(v) - c\}.$$

Clearly A is nonempty and convex. We show that A and $\operatorname{int}(\operatorname{epi} f)$ are disjoint. Indeed if by contradiction $(a, v) \in A \cap \operatorname{int}(\operatorname{epi} f)$ then

$$f(v) < a \leq \langle \xi, v\rangle - g(v) - c$$

and

$$c < \langle \xi, v\rangle - f(v) - g(v) \leq (f+g)^*(\xi) = c.$$

Then by Hahn–Banach theorem there exists $(\gamma, \zeta) \in \mathbb{R} \times V^*$ such that

$$\gamma a + \langle \zeta, v\rangle \leq \gamma b + \langle \zeta, w\rangle \tag{B.1}$$

for all $(a, v) \in \operatorname{int}(\operatorname{epi} f)$ and $(b, w) \in A$. If we fix $v = w \in \operatorname{dom} f \cap \operatorname{dom} g$, $b \in \mathbb{R}$ sufficiently small such that $(b, v) \in A$ and we let $a \to \infty$, we discover that $\gamma \leq 0$ necessarily. If $\gamma = 0$ then inequality (B.1) and Hahn–Banach theorem imply that $\operatorname{dom} f$ and $\operatorname{dom} g$ are separated by a closed hyperplane. This in turn implies that $\operatorname{int}(\operatorname{dom} f)$ and $\operatorname{dom} g$ are disjoint, but this contradicts the hypotheses so that $\beta < 0$. In this case we set $\theta = |\beta|^{-1} f\zeta$ and from inequality (B.1) we get

$$\begin{aligned} f^*(\theta) &= \sup\{\langle \theta, v\rangle - f(v) \mid v \in V\} = \\ &= \sup\{\langle \theta, v\rangle - a \mid (a, v) \in \operatorname{epi} f\} \leq \\ &\leq \inf\{\langle \theta, v\rangle - a \mid (a, v) \in A\} = \\ &= c + \inf\{\langle \theta - \xi, v\rangle + g(v) \mid v \in V\} = c - g^*(\xi - \theta). \end{aligned}$$

Then

$$f^* \square g^*(\xi) \leq f^*(\theta) + g^*(\xi - \theta) \leq c = (f+g)^*(\xi).$$

This concludes the proof. □

Proposition B.5 (properties of positively 1-homogeneous convex functions)**.** Let $f \colon V \to (-\infty, \infty]$ be a proper, l.s.c., positively 1-homogeneous and convex function. Then

(i) $f(0) = 0$ and f is subadditive
(ii) for each $x \in \operatorname{dom} f$ it holds $\partial f(x) = \{\xi \in \partial f(0) \mid f(x) = \langle \xi, x\rangle\}$
(iii) $f^* = \iota_K$ with $K = \partial f(0)$ and $f = \sigma_K$.

Proof. Step 1. Let $v \in \operatorname{dom} f$. Since f is l.s.c. and is positive 1-homogeneous

$$f(0) \leq \liminf_{n\to\infty} f\left(\frac{1}{n} v\right) = \liminf_{n\to\infty} \frac{f(v)}{n} = 0.$$

By positively 1-homogeneity and convexity we have

$$\frac{1}{2} f(v_0 + v_1) = f\left(\frac{v_0 + v_1}{2}\right) \leq \frac{f(v_0) + f(v_1)}{2}.$$

This proves (i).

Step 2. We prove that for all $\xi \in V^*$ it holds that $f^*(\xi) \in \{0, \infty\}$. Indeed let $\xi \in V^*$ such that $f^*(\xi) < \infty$. Then since f is positively 1-homogeneous, it holds $f^*(\xi) = 2^{-1} f^*(\xi)$ and this implies $f^*(\xi) = 0$.

Step 3. Let $v \in \operatorname{dom} f$ and $\xi \in \partial f(v)$. This is equivalent to the equality $f(v) + f^*(\xi) = \langle \xi, v\rangle$ which in turn implies that $f^*(\xi) < \infty$ and by previous step $f^*(\xi) = 0$. Then $f(v) = \langle \xi, v\rangle$ and for all $w \in V$

$$f(w) \geq f(v) + \langle \xi, w - v\rangle = \langle \xi, w\rangle.$$

This is equivalent to $\xi \in \partial f(0)$. We have proved that if $\xi \in \partial f(v)$ then $f(v) = \langle \xi, v\rangle$ and $v \in \partial f(0)$. It is immediate to check that also the converse is true. This proves (ii).

Step 4. It holds that $\xi \in K$ if and only if $f^*(\xi) = f(0) + f^*(\xi) = \langle \xi, 0\rangle = 0$. This proves that $f^* = \iota_K$. Since f is l.s.c. then $f = f^* * = \iota_K^* = \sigma_K$. This establishes (iii). □

B.4. **Bochner spaces.**

Definition B.6. Let V be a Banach space. We say that $u\colon [0,T] \to V$ is absolutely continuous and we write $u \in AC([0,T],V)$ if there exists $m \in L^1((0,T),(0,\infty))$ such that

$$\|u(t) - u(s)\| \leq \int_0^T m(r)dr \quad \text{for all } 0 \leq s < t \leq T.$$

Given $u\colon [0,T] \to V$ and $0 \leq a < b \leq T$ we define the total variation on $[a,b]$

$$\mathrm{Var}(u,[a,b]) = \sup \left\{ \sum_{k=1}^N \|u(t_k) - u(t_{k-1})\| \;\middle|\; a = t_0 < \cdots < t_N = b \right\}.$$

We say that u has bounded variation and we write $u \in BV([0,T],V)$ if $\mathrm{Var}(u,[0,T]) < \infty$.

Proposition B.7. Let V be a Banach space. Then $W^{1,1}((0,T),V) \subset AC([0,T],V)$. If V is separable and reflexive the equality holds.

Proof. Let $u \in W^{1,1}((0,T),V)$. Then

$$\|u(t) - u(s)\| = \left\| \int_s^t \dot{u}(r)dr \right\| \leq \int_s^t \|\dot{u}(r)\|dr.$$

This proves that $u \in AC([0,T],V)$ with $m = \|\dot{u}\|$.

Now suppose that V is reflexive. We have that

$$\left\| \frac{u(t+h) - u(t)}{h} \right\| \leq \frac{1}{h} \int_t^{t+h} m(r)dr.$$

By the Lebesgue differentiation theorem the left hand side is bounded as $h \to 0$ for a.e. $t \in (0,T)$. Then for each $h_n \to 0$, up to extracting a subsequence,

$$\frac{u(t+h_n) - u(t)}{h_n} \rightharpoonup v \quad \text{for some } v \in V.$$

We prove that v is common to each sequence. Since V^* is separable and reflexive then there exists ξ_i dense in V^*. Now for each i the map $t \mapsto \langle \xi_i, u(t)\rangle$ is $AC([0,T])$. Then for a.e. $t \in (0,T)$

$$\lim_{h\to 0} \left\langle \xi_i, \frac{u(t+h) - u(t)}{h} \right\rangle \quad \text{exists for all } i. \tag{B.2}$$

Then if $h_n \to 0$ and $k_n \to 0$ are such that

$$\frac{u(t+h_n) - u(t)}{h_n} \rightharpoonup v \quad \text{and} \quad \frac{u(t+k_n) - u(t)}{k_n} \rightharpoonup w$$

from (B.2) it follows that $\langle \xi_i, v - w\rangle = 0$ for all i. This implies $v = w$. Up to now we have proved that there exists $v\colon (0,T) \to V$ such that

$$\frac{u(t+h) - u(t)}{h} \rightharpoonup v(t) \quad \text{for a.e. } t \in (0,T).$$

Now one should prove that this convergence is actually strong. This will show automatically that $u \in W^{1,1}((0,T),V)$. □

Appendix C. Balanced-viscosity solutions

C.1. **Introduction.** Let

(V) $\qquad$ V a separable and reflexive Banach space.

Consider an evolution $u\colon [0,T] \to V$ and the doubly nonlinear equation for the evolution

$$\partial_u \mathcal{E}(u(t),t) + \partial\Phi(\dot{u}(t)) \ni 0 \quad \text{for } t \in (0,T). \tag{C.1}$$

Here $\mathcal{E}$ is the energy and Φ is the dissipation potential. In application usually $\mathcal{E}(u,t) = \Psi(u) - \langle \mathcal{F}(t), u\rangle$ where Ψ is the free energy and $\mathcal{F}$ is the external action. For a given $\eta > 0$ consider the time-rescaled evolution $u\colon (0,\eta^{-1}T) \to V$ and the time-rescaled equation

$$\partial_u \mathcal{E}(u(t),\eta t) + \partial\Phi(\dot{u}(t)) \ni 0 \quad \text{for } t \in (0,\eta^{-1}T). \tag{C.2}$$

For $\eta \to 0$ this describe the situation in which the external actions are mush slower with respect to the characteristic relaxation time of the system. Equation (C.2) can be written as

$$\partial_u \mathcal{E}(u_\eta(t),t) + \partial\Phi_\eta(\dot{u}_\eta(t)) \ni 0 \quad \text{for } t \in (0,T) \tag{C.3}$$

where $u_\eta(t) = u(\eta^{-1}t)$ and $\Phi_\eta(v) = \eta^{-1}\Phi(\eta v)$.

C.2. **Assumptions.** The energy $\mathcal{E}$ is supposed to satisfy the following assumptions.

($\mathcal{E}$.1) $\qquad$ $\mathcal{E} \in C^1(V \times [0,T])$

($\mathcal{E}$.2) $\qquad$ $\mathcal{E}(\cdot,t)$ has compact sublevels for all $t \in [0,T]$

($\mathcal{E}$.3) $\qquad$ there exists $C > 0$ such that $|\partial_t \mathcal{E}(u,t)| \leq C(1 + \mathcal{E}(u,t))$ for all $u \in V$ and $t \in [0,T]$

The dissipation potential is defined by

$$\Phi(v) = \mathcal{R}(v) + \mathcal{V}(v)$$

and for $\eta > 0$ its rescaled version is

$$\Phi_\eta(v) = \eta^{-1}\Phi_1(\eta v) = \mathcal{R}(v) + \eta^{-1}\mathcal{V}(\eta v) = \mathcal{R}(v) + \mathcal{V}_\eta(v).$$

Here $\mathcal{R}$ is a rate-independent dissipation potential and $\mathcal{V}$ is a rate-dependent dissipation potential that models viscosity. It is assumed that $\mathcal{R}$ and $\mathcal{V}$ satisfy the following assumptions.

(Φ.1) $\qquad$ $\mathcal{R}$ and $\mathcal{V}$ from V into $[0,\infty)$ are continuous, convex and strictly positive in $V \setminus \{0\}$

From Proposition B.5 it holds $\mathcal{R}(0) = 0$.

(Φ.2) $\qquad$ $\mathcal{R}$ is positively 1-homogeneous

(Φ.3) $\qquad$ $\mathcal{V}(v) = \phi(\|v\|)$ with $\phi \in C^1([0,\infty))$ convex such that $\phi(0) = 0$, $\phi'(0) = 0$, $\phi(r) > 0$ for $r > 0$ and $\lim_{r\to\infty} \phi'(r) = \infty$

We can extend ϕ to an even $C^1(\mathbb{R})$ function.

C.3. **Preliminary results regarding the dissipation potential.** Since Φ_1 is convex and $\Phi_1(0) = 0$ then for all $v \in V$ the functions $\eta \mapsto \Phi_\eta(v)$ are increasing. Then it is well defined

$$\Phi_0(v) = \lim_{\eta \to 0^+} \Phi_\eta(v) = \inf_{\eta > 0} \Phi_\eta(v) = \mathcal{R}(v).$$

Last inequality holds true since $\phi(0) = 0$ and $\phi'(0) = 0$.

Let $K = \partial\mathcal{R}(0)$. By Proposition B.5 it follows that $\mathcal{R} = \sigma_K$ and $\mathcal{R}^* = \iota_K$. It simply to check that $\Phi^*_\eta(\xi) = \eta^{-1}\Phi^*(\xi)$. It is possible to obtain a more explicit representation formula for Φ^*. By the inf-convolution duality formula

$$\Phi^*_\eta(\xi) = (\mathcal{R} + \mathcal{V}_\eta)^*(\xi) = \inf_{\zeta \in V} \left(\mathcal{R}^*(\zeta) + \mathcal{V}^*_\eta(\xi - \zeta)\right) = \eta^{-1} \inf_{\zeta \in K} \mathcal{V}^*(\xi - \zeta) = \eta^{-1} \min_{\zeta \in K} \mathcal{V}^*(\xi - \zeta). \tag{C.4}$$

Last equality holds true because, since K is closed and convex, it is also weakly closed, $\mathcal{V}$ is convex and coercive, then the infimum is actually a minimum. Now $\mathcal{V}^*(\xi) = \phi^*(\|\xi\|)$.

Lemma C.1. *The function ϕ^* satisfies $\phi^*(0) = 0$, $(\phi^*)'(0) = 0$ and is monotone.*

Proof. Let $r = \rho(\gamma)$ the solution to $\gamma - \phi'(\rho(\gamma)) = 0$. Clearly $\rho \in C^1(\mathbb{R})$ and $\rho(0) = 0$. Then $\phi^*(\gamma) = \gamma\rho(\gamma) - \phi(\rho(\gamma))$. Now $\phi^*(0) = -\phi(0) = 0$ and $(\phi^*)'(0) = -\phi'(0) = 0$. □

Using that lemma and (C.4)

$$\Phi^*_\eta(\xi) = \eta^{-1} \min_{\zeta \in K} \phi^*(\|\xi - \zeta\|). \tag{C.5}$$

C.4. **Absolutely continuous and BV evolutions.** Let $\mathcal{R} \colon V \to [0, \infty)$ satisfying (Φ.1) and (Φ.2). We say that a map $u \colon [0, T] \to V$ is $\mathrm{AC}_\mathcal{R}([0, T], V)$ if there exists $m \in L^1((0, T), [0, \infty))$ such that

$$\mathcal{R}(u(t_1) - u(t_0)) \leq \int_{t_0}^{t_1} m(s)ds \quad \text{for every } 0 \leq t_0 < t_1 \leq T. \tag{C.6}$$

Lemma C.2. *Let $u \in AC_\mathcal{R}([0, T], V)$. Then*

$$\mathcal{R}[\dot{u}](t) \doteq \lim_{h \to 0} \mathcal{R}\left(\frac{u(t + h) - u(t)}{h}\right)$$

exists for a.e. $t \in (0, T)$ and $\mathcal{R}[\dot{u}] \in L^1((0, T))$. Moreover for all m as in (C.6)

$$\mathcal{R}[\dot{u}](t) \leq m(t) \quad \textit{for a.e. } t \in (0, T).$$

Proof. For any $s \in (0, T)$ and $t \in (0, T)$ let $\delta_s(t) = \mathcal{R}(u(t) - u(s))$. Then

$$\delta_s(t_1) - \delta_s(t_0) \leq \mathcal{R}(u(t_1) - u(t_0)) \leq \int_{t_0}^{t_1} m(s)ds \quad \text{for all } 0 \leq t_0 < t_1 \leq T.$$

The second inequality holds since $\mathcal{R}$ is sub additive by Proposition B.5 so

$$\delta_s(t_1) = \mathcal{R}(u(t_1) - u(s)) \leq \mathcal{R}(u(t_1) - u(t_0)) + \mathcal{R}(u(t_0) - u(s)) = \mathcal{R}(u(t_1) - u(t_0)) + \delta_s(t_0).$$

Then the map $t \mapsto \delta_s(t) - \int_0^t m(r)dr = \beta_s(t)$ is non increasing on $(0, T)$ and in particular it is differentiable a.e. on $(0, T)$. This in turns implies that δ_s is differentiable a.e. on $(0, T)$ and

$$\dot{\delta}_s(t) \leq \delta(t) \doteq \liminf_{h \to 0^+} \mathcal{R}\left(\frac{u(t + h) - u(t)}{h}\right) \quad \text{for a.e. } t \in (0, T). \tag{C.7}$$

Note that

$$0 \leq \delta(t) \leq \liminf_{h \to 0^+} \frac{1}{h} \int_t^{t+h} m(s)ds = m(t) \quad \text{for a.e. } t \in (0, T) \tag{C.8}$$

and this implies $\delta \in L^1((0,T))$. Since $\delta_s(t) = \beta_s(t) + \int_0^t m(r)dr$ and β_s is non increasing then the singular part of the distributional derivative of δ_s is non positive and

$$\mathcal{R}(u(t) - u(s)) = \delta_s(t) \le \int_s^t \dot{\delta}_s(r)dr \le \int_s^t \delta(r)dr. \tag{C.9}$$

Let

$$\gamma(t) \doteq \limsup_{h\to 0^+} \mathcal{R}\left(\frac{u(t+h)-u(t)}{h}\right).$$

By (C.9) then $\delta(t) \le \gamma(t) \le \delta(t)$. This proves that

$$\delta^+(t) \doteq \lim_{h\to 0^+} \mathcal{R}\left(\frac{u(t+h)-u(t)}{h}\right) = \delta(t) \quad \text{exists for a.e. } t \in (0,T).$$

Applying the previous argument to $t \mapsto u(T-t)$ and $v \mapsto \mathcal{R}(-v)$ in place of u and $\mathcal{R}$ respectively, follows that

$$\delta^-(t) = \lim_{h\to 0^-} \mathcal{R}\left(\frac{u(t+h)-u(t)}{h}\right) = \lim_{h\to 0^+} \mathcal{R}\left(\frac{u(t)-u(t-h)}{h}\right) \quad \text{exists for a.e. } t \in (0,T)$$

and satisfy the conditions

$$\mathcal{R}(u(t) - u(s)) \le \int_s^t \delta^-(r)dr. \tag{C.10}$$

and $\delta^-(t) \le m(t)$ for a.e. $t \in (0,T)$. This means that δ^+ and δ^- coincides both with the minimal $m \in L^1((0,T),(0,\infty))$ for which (C.6) holds. ☐

Given a map $u\colon [0,T] \to V$ we define the total variation on $[a,b] \subset [0,T]$ as

$$\mathrm{Var}_{\mathcal{R}}(u,[a,b]) = \sup\left\{ \sum_{k=1}^N \mathcal{R}(u(t_k)) - \mathcal{R}(u(t_{k-1})) \;\middle|\; a = t_0 < \cdots < t_N = b \right\}.$$

We say that $u \in \mathrm{BV}_{\mathcal{R}}([0,T],V)$ if $\mathrm{Var}_{\mathcal{R}}(u,[0,T]) < \infty$.

Lemma C.3. *It holds that $BV([0,T],V) \subset BV_{\mathcal{R}}([0,T],V)$.*

Proof. Step 1. We first prove that there exists $C > 0$ such that $\mathcal{R}(v) \le C\|v\|$. Indeed by contradiction suppose that there exists a sequence such that $\mathcal{R}(v_k) \ge k\|v_k\|$. By 1-homogeneity we can suppose that $\|v_k\| = 1$ without loss of generality. Moreover up to extracting a subsequence $v_k \rightharpoonup v$.

My proof is wrong, but in the case $V = L^2(\Omega)$ with Ω bounded and $\mathcal{R} = \|\cdot\|_1$ for example the result is trivially true.

Step 2. By previous step $\mathrm{Var}_{\mathcal{R}}([0,T],V) \le C\,\mathrm{Var}([0,T],V)$. This completes the proof. ☐

C.5. **A chain rule formula for the energy functional.**

Proposition C.1. Let $u \in AC([0,T],V)$. Then $t \mapsto \mathcal{E}(u(t),t)$ is absolutely continuous in $[0,T]$ and

$$\frac{d}{dt}\mathcal{E}(u(t),t) = \langle \partial_u \mathcal{E}(u(t),t), \dot{u}(t)\rangle + \partial_t \mathcal{E}(u(t),t) \quad \text{for a.e. } t \in (0,T). \tag{C.11}$$

Proof. Step 1. Since u is continuous then $K = u([0,T])$ is compact. Indeed let $u_n = u(t_n)$ be a sequence in K. Then up to a subsequence $t_n \to t$ and by continuity $u_n = u(t_n) \to u(t)$.

Step 2. Since $\mathcal{E} \in C^1(V \times [0,T])$ then by Proposition B.2 it is Lipschitz in $K \times [0,T]$. Let $L > 0$ be the Lipschitz constant.

Step 3. Let $0 \le s < t \le T$ and $m \in L^1((0,T),(0,\infty))$ as in Definition B.6. Then

$$|\mathcal{E}(u(t),t) - \mathcal{E}(u(s),s)| \le L\left(\|u(t)-u(s)\| + |t-s|\right) \le L\int_s^t (m(r)+r)\,dr.$$

This proves that $t \mapsto \mathcal{E}(u(t),t)$ is absolutely continuous in $[0,T]$.

Step 4. By Proposition B.7 it follows that u is a.e. differentiable in $(0,T)$. This implies formula (C.11). □

C.6. **Existence results in presence of positive viscosity.**

Proposition C.2. Let $u_0 \in V$. Then there exists $u_\eta \in \mathrm{AC}([0,T],V)$ such that $u_\eta(0) = u_0$ and

$$\partial_u \mathcal{E}(u_\eta(t),t) + \partial\Phi_\eta(\dot u_\eta(t)) \ni 0 \quad \text{for a.e. } t \in (0,T).$$

Moreover the identity

$$\text{(C.12)} \quad \mathcal{E}(u_\eta(t),t) + \int_s^t \left(\Phi_\eta(\dot u_\eta(r)) + \Phi_\eta^*(-\partial_u\mathcal{E}(u_\eta(r),r)\right)dr = \mathcal{E}(u_\eta(s),s) + \int_s^t \partial_t\mathcal{E}(u_\eta(r),r)dr$$

holds for every $0 \le s < t \le T$.

Proof. I still have to report the proof of existence. We prove the identity (C.12). By the properties of subdifferentials

$$\Phi_\eta(\dot u(t)) + \Phi_\eta^*(-\partial_t\mathcal{E}(u(t),t)) = -\langle \partial_u\mathcal{E}(u(t),t), \dot u(t)\rangle \quad \text{for a.e. } t \in (0,T).$$

By (C.11) we have

$$\Phi_\eta(\dot u(t)) + \Phi_\eta^*(-\partial_t\mathcal{E}(u(t),t)) = -\frac{d}{dt}\mathcal{E}(u(t),t) + \partial_t\mathcal{E}(u(t),t) \quad \text{for a.e. } t \in (0,T).$$

Integrating this identity in (s,t) we get the thesis. □

C.7. **Contact dissipation potential.** The dissipation contact potential $\mathfrak{p}\colon V \times V^* \to [0,\infty)$ is

$$\mathfrak{p}(v,\xi) = \inf_{\eta>0}\left(\Phi_\eta(v) + \Phi_\eta^*(\xi)\right) = \inf_{\eta>0} \eta^{-1}\left(\Phi(\eta v) + \Phi^*(\xi)\right).$$

By (C.5)

$$\begin{aligned}
\mathfrak{p}(v,\xi) &= \inf_{\eta>0}\left(\mathcal{R}(v) + \eta^{-1}\phi(\eta\|v\|) + \eta^{-1}\min_{\zeta\in K}\phi^*(\|\xi-\zeta\|)\right) = \\
&= \mathcal{R}(v) + \inf_{\eta>0}\left[\eta^{-1}\min_{\zeta\in K}\left(\phi(\eta\|v\|) + \phi^*(\|\xi-\zeta\|)\right)\right] = \\
&= \mathcal{R}(v) + \min_{\zeta\in K}\inf_{\eta>0}\frac{\phi(\eta\|v\|) + \phi^*(\|\xi-\zeta\|)}{\eta}
\end{aligned}$$

where last equality holds because the minimum point does not depends on η. But for all r and γ in $\mathbb{R}$

$$\inf_{\eta>0}\frac{\phi(\eta r) + \phi^*(\gamma)}{\eta} = \gamma r.$$

Indeed by Young–Fenchel inequality

$$\frac{\phi(\eta r) + \phi^*(\gamma)}{\eta} \ge \gamma r$$

and the identity holds for η such that $\gamma = \phi'(\eta r)$. Then

$$\mathfrak{p}(v, \xi) = \mathcal{R}(v) + \|v\| \min_{\zeta \in K} \|\xi - \zeta\|. \tag{C.13}$$

Let $\mathfrak{f} \colon V \times V \times [0, T] \to [0, \infty)$ defined by

$$\mathfrak{f}(u, v, t) = \mathfrak{p}(v, -\partial_u \mathcal{E}(u, t)) = \mathcal{R}(v) + \mathfrak{e}(u, t)\|v\| \tag{C.14}$$

where $\mathfrak{e} \colon V \times [0, T] \to [0, \infty)$ is defined by

$$\mathfrak{e}(u, t) = \min_{\zeta \in K} \|\partial_u \mathcal{E}(u, t) + \zeta\|.$$

Notice that $\mathfrak{e}(u, t) = 0$ if and only if $-\partial_u \mathcal{E}(u, t) \in K$.

C.8. **Parametrized balanced-viscosity solutions.** Let $u_\eta \in \mathrm{AC}([0, T], V)$ be the solution to the doubly nonlinear equation

$$\partial_u \mathcal{E}(u_\eta(t), t) + \partial \Phi_\eta(\dot{u}_\eta(t)) \ni 0 \quad \text{for } t \in (0, T)$$

The identity (C.12) implies

$$\mathcal{E}(u_\eta(t), t) + \int_s^t \mathfrak{f}(u_\eta(t), \dot{u}_\eta(t), t) dr \le \mathcal{E}(u_\eta(s), s) + \int_s^t \partial_t \mathcal{E}(u_\eta(t), t) dr \tag{C.15}$$

for all $0 \le s < t \le T$.

Lemma C.4. *There exists a constant $C > 0$ such that*

$$\int_0^T \mathfrak{f}(u_\eta(t), \dot{u}_\eta(t), t) dt \le C \quad \text{for all } \eta > 0.$$

Proof. Since $\mathfrak{f} \ge 0$ by $(\mathcal{E}.3)$ then inequality (C.15) implies

$$\mathcal{E}(u_\eta(t), t) \le \mathcal{E}(u_0, 0) + \int_0^t \partial_t \mathcal{E}(u(s), s) ds \le \mathcal{E}(u_0, 0) + C \int_0^t (1 + \mathcal{E}(u_\eta(s), s)) ds.$$

By $(\mathcal{E}.3)$ and Gronwall inequality $-1 \le \mathcal{E}(u_\eta(t), t) \le C_1$ for all $t \in [0, T]$ and $\eta > 0$. Then (C.15) implies

$$\int_0^T \mathfrak{f}(u_\eta(t), \dot{u}_\eta(t), t) dt \le 1 + \mathcal{E}(u_0, 0) + CT(1 + C_1).$$

This concludes the proof. □

Let $\boldsymbol{s}_\eta \colon [0, T] \to [0, S_\eta]$ defined by

$$\boldsymbol{s}_\eta(t) = t + \int_0^t \mathfrak{f}(u_\eta(s), \dot{u}_\eta(s), s) ds \quad \text{and} \quad S_\eta = \boldsymbol{s}_\eta(T). \tag{C.16}$$

Let also

$$\boldsymbol{t}_\eta = \boldsymbol{s}_\eta^{-1} \colon [0, S_\eta] \to [0, T] \quad \text{and} \quad \boldsymbol{u}_\eta = u_\eta(\boldsymbol{t}_\eta) \colon [0, S_\eta] \to V.$$

To write the time-rescaled energy identity we introduce the map $\mathfrak{P}_\eta \colon (0, \infty) \times V \times \mathbb{R} \times V^* \to [0, \infty]$ defined by

$$\mathfrak{P}_\eta(a, v; \gamma, \xi) = a\Phi_\eta(a^{-1} v) + a\Phi_\eta^*(\xi) + a\gamma$$

and $\mathfrak{F}_\eta \colon [0, T] \times V \times [0, \infty] \times V \to [0, \infty]$ defined by

$$\mathfrak{F}_\eta(t, u; a, v) = \mathfrak{P}_\eta(a, v; -\partial_t \mathcal{E}(u, t), -\partial_u \mathcal{E}(u, t))$$

Then (C.12) becomes

$$\mathcal{E}(\boldsymbol{u}_\eta(t),\boldsymbol{t}_\eta(t)) + \int_s^t \mathfrak{F}_\eta(\boldsymbol{t}_\eta(r),\boldsymbol{u}_\eta(r);\dot{\boldsymbol{t}}_\eta(r),\dot{\boldsymbol{u}}_\eta(r))dr = \mathcal{E}(\boldsymbol{u}_\eta(s),\boldsymbol{t}_\eta(s)) \quad \text{for all } 0 \le s < t \le S_\eta.$$

Identity (C.16) implies

$$1 = \dot{\boldsymbol{t}}_\eta(s) + \mathfrak{p}(\boldsymbol{u}_\eta(s),\dot{\boldsymbol{u}}_\eta(s),\boldsymbol{t}_\eta(s)) \quad \text{for a.e. } s \in (0,S_\eta).$$

Now let $\mathfrak{P}\colon [0,\infty)\times V\times\mathbb{R}\times V^* \to [0,\infty]$ defined by

$$\mathfrak{P}(a,v;\gamma,\xi) = \begin{cases} \mathcal{R}(v) + \iota_K(\xi) + a\gamma & \text{if } a > 0 \\ \mathfrak{p}(v,\xi) & \text{if } a = 0. \end{cases}$$

and $\mathfrak{F}\colon [0,T]\times V\times[0,\infty)\times V \to [0,\infty]$ defined by

$$\mathfrak{F}(t,v;a,u) = \mathfrak{P}(a,v;-\partial_t\mathcal{E}(u,t),-\partial_u\mathcal{E}(u,t)).$$

Definition C.3. We say that a pair $(\boldsymbol{u},\boldsymbol{t})\colon [0,S]\to V\times[0,T]$ is an admissible parametrized curve if

(i) $\boldsymbol{t}$ is non decreasing and absolutely continuous, $\boldsymbol{u} \in \mathrm{AC}_{\mathcal{R}}([0,S],V)$ and $\sup_{t\in[0,S]}\boldsymbol{u}(s) < \infty$
(ii) $\boldsymbol{u}$ locally absolutely continuous (with respect to $\|\cdot\|$) in the open set

$$G = \{s\in[0,S] \mid \mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s)) > 0\} = \{s\in[0,S] \mid -\partial_u\mathcal{E}(\boldsymbol{u}(s),\boldsymbol{t}(s)) \notin K\}$$

(iii) the estimate

$$\int_0^S \mathfrak{f}[\boldsymbol{u},\dot{\boldsymbol{u}},\boldsymbol{t}](s)ds = \int_0^S \mathcal{R}[\dot{\boldsymbol{u}}](s)ds + \int_G \mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))\|\dot{\boldsymbol{u}}(s)\|ds < \infty$$

holds.

For every admissible parametrized curve and for a.e. $s\in[0,S]$ is well defined

$$\mathfrak{F}[\boldsymbol{t},\boldsymbol{u};\dot{\boldsymbol{t}},\dot{\boldsymbol{u}}](s) = \begin{cases} \mathcal{R}[\dot{\boldsymbol{u}}](s) + \iota_K(-\partial_u\mathcal{E}(\boldsymbol{u}(s),\boldsymbol{t}(s))) - \dot{\boldsymbol{t}}(s)\partial_t\mathcal{E}(\boldsymbol{u}(t),\boldsymbol{t}(s)) & \text{if } \dot{\boldsymbol{t}}(s) > 0 \\ \mathcal{R}[\dot{\boldsymbol{u}}](s) + \mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))\|\dot{\boldsymbol{u}}(s)\| & \text{if } \boldsymbol{t}(s) = 0. \end{cases}$$

where is implicit that if $s\notin G$ then $\mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))\|\dot{\boldsymbol{u}}(s)\| = 0$.

We denote with $\mathsf{A}([0,S],V\times[0,T])$ the set of all admissible parametrized curves and we say that $(\boldsymbol{u},\boldsymbol{t})$ is

(i) non degenerate if $\dot{\boldsymbol{t}}(s) + \mathcal{R}[\dot{\boldsymbol{u}}](s)$ for a.e. $s\in[0,S]$
- COMPLETE!!!

C.9. **An example.** Let $V = L^2(\Omega)$ with $\Omega\subset\mathbb{R}^d$ open and bounded. Let

$$\mathcal{R}(v) = \int_\Omega |v|dx \quad \text{and} \quad \mathcal{V}(v) = \frac{1}{2}\int_\Omega |v|^2dx.$$

They satisfy the assumptions (D). In particular the continuity of $\mathcal{R}$ follows from Hölder inequality

$$|\mathcal{R}(v)-\mathcal{R}(w)| \le \int_\Omega \left||v|-|w|\right| dx \le \int_\Omega |v-w|dx \le C\|v-w\|_2.$$

It holds that K is the closed unit ball in $L^\infty(\Omega)$. Indeed if $\xi\in V^* = L^2(\Omega)$ satisfy $\|\xi\|_\infty \le 1$ then

$$\mathcal{R}^*(\xi) = \sup_{v\in V}\int_\Omega (\xi v - |v|)\,dx \le \sup_{v\in V}\int_\Omega (\xi v - |v|)\,dx \le \sup_{v\in V}\int_\Omega (|\xi|-1)|v|dx \le 0$$

and $\mathcal{R}^*(\xi) \geq 0$ trivially. If $\|\xi\|_\infty > 1$ then there exists $\epsilon > 0$ and $w \in L^1(\Omega)$ such that $\langle \xi, w\rangle \geq 1+\epsilon$. Since $L^2(\Omega)$ is dense in $L^1(\Omega)$ then for all $\delta > 0$ there exist $v \in L^2(\Omega)$ such that $\|w - v\| \leq \delta$. Then for all $\lambda > 0$

$$\mathcal{R}^*(\xi) \geq \langle \xi, \lambda v\rangle - \|\lambda v\|_1 \geq \lambda\left(\langle \xi, w\rangle - \|w\|_1 - \delta(1 + \|\xi\|)\right) \geq \lambda\left(\epsilon - \delta(1 + \|\xi\|)\right).$$

Choosing δ such that $\epsilon - \delta(1 + \|\xi\|) > 0$ and letting $\lambda \to \infty$ we get $\mathcal{R}(\xi) = \infty$. Since $\mathcal{V}^* = \frac{1}{2}\|\cdot\|_2^2$ and by (C.4)

$$\Phi_\epsilon^*(\xi) = \frac{1}{2\eta}\min_{\zeta\in K}\|\xi - \zeta\|_2^2 = \frac{1}{2\eta}\int_\Omega \min_{|\zeta|\leq 1}|\xi - \zeta|^2 dx.$$

By (C.13) it follows

$$\mathfrak{p}(v, \xi) = \|v\|_1 + \|v\|_2 \min_{\zeta\in K}\|\xi - \zeta\|_2.$$

Appendix D. Energetic solutions

Email address: `bernardo.ardini@studenti.unipd.it`